Probability and Stochastic Processes

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Chapter 4 Sequences of Random Variables

In this chapter, we use the theory of convergence of sequences of random variables to prove the law of large numbers and the central limit theorem.

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1. Family of Distribution Functions and Numerical Characteristics
2. Modes of Convergence
3. The Law of Large Numbers
4. The Central Limit Theorem

Family of Distribution Functions and Numerical Characteristics

In this chapter, we use the theory of convergence of sequences of random variables to prove the law of large numbers and the central limit theorem.

Family of Distribution Functions and Numerical Characteristics

Let Ω be a sample space. We say that $\{X_n(\omega), \omega \in \Omega\}_{n=1}^{\infty}$ is a ***sequence of random variables*** defined on the sample space Ω if all the random variables X_n belonging to the sequence $\{X_n\}_{n=1}^{\infty}$ are functions from Ω to $\mathbb{R}$. We usually denote the sequence of random variables by $\{X_n\}_{n=1}^{\infty}$.

Family of Distribution Functions and Numerical Characteristics

Example

Suppose that a die is rolled and the number is observed.
(i) Let X_n denote the number that appears at the nth ($n \geqslant 1$) time. For $n = 1, 2, \ldots,$ X_n denotes different random variable, thus $\{X_n\}_{n=1}^{\infty}$ is a sequence of random variables. We call it Bernoulli sequence.
(ii) Let Y_n denote the maximum number that appears in the first n times. $\{Y_n\}_{n=1}^{\infty}$ is also a sequence of random variables. $\square$

Family of Distribution Functions and Numerical Characteristics

Example

Consider a farmer using an old tractor. The tractor is often in the repair shop but it always takes only one day to get it running again. The first day out of the shop it always works but on any given day thereafter, independent of its previous history, there is a 10% chance of it not working and thus being sent back to the shop. Let $X_0, X_1, \cdots$ be random variables denoting the daily condition of the tractor, where a one denotes the working condition and a zero denotes the failed condition. In other words, $X_n = 1$ denotes that the tractor is working on day n and $X_n = 0$ denotes it being in the repair shop on day n. Thus, $X_0, X_1, \cdots$ is a sequence of random variables. $\square$

Family of Distribution Functions and Numerical Characteristics

How to describe a sequence by distributions?

Notice that for a specific n, X_n is a r.v.. The distribution function of this r.v. will, in general, depend on n. We have

$$F(x; n) = P(X_n \leqslant x).$$

The function $F(x; n)$ sometimes denoted by $F_n(x)$, which is called the first-order distribution function of the sequence $\{X_n\}_{n=1}^{\infty}$. $\{F_n(x), n = 1, 2, \cdots\}$ will be called the family of the first-order distribution functions.

Family of Distribution Functions and Numerical Characteristics

Similarly, for two specific n and m, $n \neq m$, X_n and X_m are two random variables. The joint distribution function of them will, in general, depend on n and m. Denoting it by $F(x, y; n, m)$, we have

$$F(x, y; n, m) = P(X_n \leqslant x, X_m \leqslant y).$$

We call the function $F(x, y; n, m)$ the second-order distribution function of the sequence $\{X_n\}_{n=1}^{\infty}$, and $\{F(x, y; n, m), n, m = 1, 2, \cdots \text{and } n \neq m\}$ will the family of the second-order distribution functions.

Family of Distribution Functions and Numerical Characteristics

For all positive integers $k \geq 1$, and for all times $n_1, n_2, \cdots, n_k = 1, 2, \cdots$, consider the random vector $(X_{n_1}, X_{n_2}, \cdots, X_{n_k})$. The kth-order probability distribution function of the sequence $\{X_n\}_{n=1}^{\infty}$ is defined by

$$\begin{aligned} & F(x_1, x_2, \cdots, x_k; n_1, n_2, \cdots, n_k) \\ = & P(X(t_1) \leq x_1, X(t_2) \leq x_2, \cdots, X(t_n) \leq x_n). \end{aligned}$$

For a fixed k,

$$\{F(x_1, x_2, \cdots, x_k; n_1, n_2, \cdots, n_k),\ n_i = 1, 2, \cdots, i = 1, 2, \cdots, k\}$$

will be called the family of the kth-order distribution functions.

Family of Distribution Functions and Numerical Characteristics

Definition

Suppose that $\{X_n\}_{n=1}^{\infty}$ *is a random sequence. The* ***family of distribution functions*** *of* $\{X_n\}_{n=1}^{\infty}$ *is defined by*

$$\{F(x_1, x_2, \cdots, x_k; n_1, n_2, \cdots, n_k),$$

$$\forall\ k,\ \forall\ n_i = 1, 2, \cdots, i = 1, 2, \cdots, k\}.$$

Family of Distribution Functions and Numerical Characteristics

Now we introduce some terminology related to sequences of random variables.
Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of random variables defined on a sample space Ω. We say that $\{X_n\}_{n=1}^{\infty}$ is an ***independent sequence of random variables*** (or a sequence of independent random variables) if every finite subset of $\{X_n\}_{n=1}^{\infty}$(i.e., every finite set of random variables belonging to the sequence) is a set of mutually independent random variables. We say that $\{X_n\}_{n=1}^{\infty}$ is a ***sequence of identically distributed random variables*** if any two elements of the sequence have the same distribution function:

$$\forall x \in \mathbb{R},\ \forall i, j \in \mathbb{N},\ F_i(x) = F_j(x).$$

Family of Distribution Functions and Numerical Characteristics

We say that $\{X_n\}_{n=1}^{\infty}$ is a ***sequence of independent and identically distributed random variables*** (or an IID sequence of random variables), if $\{X_n\}_{n=1}^{\infty}$ is both a sequence of independent random variables and a sequence of identically distributed random variables.

Family of Distribution Functions and Numerical Characteristics

Take a first group of k successive terms of the sequence $X_{n+1}, \cdots, X_{n+k}$, and take a second group of k successive terms of the sequence $X_{n+m+1}, \cdots, X_{n+m+k}$. The second group is located m positions after the first group. The sequence is said to be ***strictly stationary*** if and only if:

$$F(x_1, \cdots, x_k; n+1, \cdots, n+k) = F(x_1, \cdots, x_k; n+m+1, \cdots, n+m+k)$$

for any $n, m, k \in \mathbb{N}$ and for any vector $(x_1, \cdots x_k) \in \mathbb{R}^k$.

Family of Distribution Functions and Numerical Characteristics

In other words, a sequence is strictly stationary if and only if the two random vectors $(X_{n+1}, \cdots, X_{n+k})$ and $(X_{n+m+1}, \cdots, X_{n+m+k})$ have the same distribution (for any n, m and k). Requiring strict stationarity is weaker than requiring that a sequence be IID: if a sequence is an IID sequence, then it is also strictly stationary, while the converse is not necessarily true.

Family of Distribution Functions and Numerical Characteristics

If we know all information about the family of distribution functions, then we say that we have statistically specified (statistically determined) the sequence.
For the Bernoulli sequence, for a specific n, we know

$$P(X_n = i) = 1/6, \qquad i = 1, 2, \cdots, 6.$$

For two specific n and m, $n \neq m$,

$$P(X_n = i, X_m = j) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}.$$

For a fixed k.

$$P(X_{n_1} = i_1, \cdots, X_{n_k} = i_k) = (\frac{1}{6})^k = \frac{1}{6^k},$$

That is, Bernoulli sequence is not only a strict stationary sequence but an IID sequence.

Family of Distribution Functions and Numerical Characteristics

However, in most cases, it is impossible to get all information about the family of distribution functions of a sequence. The alternative method is to use the moment functions, which play an important role in practical applications.

Family of Distribution Functions and Numerical Characteristics

Assume that $\{X_n, n = 1, 2, \dots\}$ is a random sequences. The first moment or mean function of $\{X_n, n = 1, 2, \dots\}$ is defined by

$$\mu_X(n) = E(X_n).$$

The power function is defined by

$$\Psi_X^2(n) = E(X_n^2).$$

The variance function is defined by

$$\sigma_X^2(n) = E\{[X_n - \mu_X(n)]^2\}.$$

The autocorrelation function is defined by

$$R_X(n, m) = E(X_n X_m).$$

The autocovariance function is defined by

$$C_X(n, m) = Cov(X_n, X_m) = E\big\{\big[X_n - \mu_X(n)\big]\big[X_m - \mu_X(m)\big]\big\}.$$

Family of Distribution Functions and Numerical Characteristics

The moments of the random sequences follows that

Property

(i) $\Psi_X^2(n) = R_X(n,n)$,
(ii) $C_X(n_1,n_2) = R_X(n_1,n_2) - \mu_X(n_1)\mu_X(n_2)$,
(iii) $\sigma_X^2(n) = C_X(n,n) = R_X(n,n) - \mu_X^2(n)$.

These concepts can be generalized to two-dimensional random sequence.

Definition

Suppose that $\{X_n\}_{n=1}^{\infty}$ *and* $\{Y_n\}_{n=1}^{\infty}$ *are two random sequences on* Ω. *If for all* n, (X_n, Y_n) *is a random vector, then we say that* $\{(X_n, Y_n), n \geqslant 1\}$ *is a two-dimensional random sequence.*

Family of Distribution Functions and Numerical Characteristics

Definition

Suppose that $\{(X_n, Y_n), n \geqslant 1\}$ *is a two-dimensional random sequence. For all positive integers* k_1 *and* k_2*, and for all* $n_1, n_2, \cdots, n_{k_1}; n'_1, n'_2, \cdots, n'_{k_2}$*, the* $(k_1 + k_2)$*th-order probability distribution function of the sequence* $\{(X_n, Y_n), n \geqslant 1\}$ *is defined by*

$$\begin{aligned} &F(x_1, \cdots, x_{k_1}; y_1, \cdots, y_{k_2}; n_1, n_2, \cdots, n_{k_1}; n'_1, n'_2, \cdots, n'_{k_2}) \\ &\quad = P(X_{n_1} \leqslant x_1, \cdots, X_{n_{k_1}} \leqslant x_{k_1}; Y_{n'_1} \leqslant y_1, \cdots, Y_{n'_{k_2}} \leqslant y_{k_2}). \end{aligned}$$

Family of Distribution Functions and Numerical Characteristics

The cross-correlation function of $\{(X_n, Y_n), n \geqslant 1\}$ is defined by

$$R_{XY}(n,m) = E(X_n Y_m).$$

The cross-covariance function of $\{(X_n, Y_n), n \geqslant 1\}$ is defined by

$$C_{XY}(n,m) = Cov(X_n, Y_m) = E\big\{\big[X_n - \mu_X(n)\big]\big[Y_m - \mu_Y(m)\big]\big\}.$$

It can be easily shown that
$C_{XY}(n,m) = R_{XY}(n,m) - \mu_X(n)\mu_Y(m).$

Family of Distribution Functions and Numerical Characteristics

The autocorrelation function is often used to describe the relationship among the elements in a sequence. In fact, the relationships among random variables differ widely in different sequences.

According to the relationships between different elements in a sequence, we can classify some sequences of random variables. This usually belongs to the category of stochastic process and will be discussed in later chapters.

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Modes of Convergence

A fundamental problem in the theory of probability is the determination of the asymptotic properties of random sequences. Whenever we want to assess whether a sequence is convergent to a limit, we need to define a distance function (or metric) to measure the distance between the terms of the sequence. Intuitively, a sequence converges to a limit if, by dropping a sufficiently high number of initial terms of the sequence, the remaining terms can be made as close to each other as we wish.

Modes of Convergence

The problem is how to define "close to each other". As we have explained, the concept of "close to each other" can be made fully rigorous using the notion of a metric. Therefore, discussing convergence of a sequence of random variables boils down to discussing what metrics can be used to measure the distance between two random variables.

Modes of Convergence

There are four principle ways of interpreting the statement "$X_n \to X$ as $n \to \infty$".

Definition

Let $\{X_n\}_{n=1}^{\infty}$ be a random sequence. We say

(a) $X_n \to X$ ***almost surely***, *written* $X_n \xrightarrow{a.s.} X$, *if* $\{\omega \in \Omega : X_n(\omega) \to X(\omega)$ *as* $n \to \infty\}$ *is an event whose probability is 1.*

(b) $X_n \to X$ ***in probability***, *written* $X_n \xrightarrow{P} X$, *if*

$$P(|X_n - X| > \varepsilon) \to 0 \text{ as } n \to \infty \text{ for all } \varepsilon > 0.$$

Modes of Convergence

Definition

(c) $X_n \to X$ ***in rth mean***, *where* $r \geqslant 1$, *written* $X_n \xrightarrow{r} X$, *if* $E|X_n^r| < \infty$ *for all* n *and*

$$E(|X_n - X|^r) \to 0 \text{ as } n \to \infty.$$

X *is called the mean-square limit of the sequence* X_n *if* $r = 2$.

(d) $X_n \to X$ ***in distribution***, *written* $X_n \xrightarrow{D} X$, *if*

$$P(X_n \leqslant x) \to P(X \leqslant x) \text{ as } n \to \infty$$

for all points x *at which the function* $F(x) = P(X \leqslant x)$ *is continuous.*

Modes of Convergence

Example

Suppose that the sample space Ω is the interval $[0, 1]$. It is possible to build a probability P on Ω, such that P assigns to each sub-interval of a probability equal to its length:

$$\text{if } 0 \leqslant a \leqslant b \leqslant 1, \text{and } E = [a, b], \text{then } P(E) = b - a.$$

It is obvious that for any $\omega \in \Omega$, we have $P(\omega) = 0$.
Now, consider a sequence of $\{X_n(\omega)\}_{i=1}^{\infty}$ random variables defined as follows:

$$X_n(\omega) = \begin{cases} 1 & \text{for } \omega = 0, \\ 1/n & \text{for } \omega \neq 0. \end{cases}$$

Prove $X_n(\omega) \to 0$ almost surely.

Modes of Convergence

Proof. Note when $\omega \in (0, 1]$, the sequence of real numbers $\{X_n(\omega)\}_{i=1}^{\infty}$ converges to 0, because

$$\lim_{n\to\infty} X_n(\omega) = \lim_{n\to\infty} \frac{1}{n} = 0.$$

However, when $\omega = 0$, the sequence of real numbers $\{X_n(\omega)\}_{i=1}^{\infty}$ is not convergent to 0, because

$$\lim_{n\to\infty} X_n(\omega) = \lim_{n\to\infty} 1 = 1.$$

Define a constant random variable $X(\omega)$ as follows:

$$X(\omega) = 0, \qquad \forall \omega \in [0, 1].$$

Modes of Convergence

We have that

$$\{\omega \in \Omega : X_n(\omega) \text{ does not converge to } X(\omega) \text{ as } n \to \infty\} = \{0\}.$$

But $P(0) = 0$, which means that the event

$$\{\omega \in \Omega : X_n(\omega) \text{ does not converge to } X(\omega) \text{ as } n \to \infty\}$$

is a zero-probability event. Therefore, the sequence $\{X_n(\omega)\}_{i=1}^{\infty}$ converges to X almost surely. $\square$

In other words, almost sure convergence requires that the sequences $\{X_n(\omega)\}_{i=1}^{\infty}$ converge for all sample points $\omega \in \Omega$, except, possibly, for a very small set, which must be included in a zero-probability event.

Modes of Convergence

Example

Let $X_n \sim N(0, 1/n)$. Prove $X_n \to 0$ in probability.

Proof. By using Chebyshev's inequality, for any $\varepsilon > 0$,

$$P(|X_n - 0| > \varepsilon) \leqslant \frac{Var(X_n)}{\varepsilon^2} = \frac{1/n}{\varepsilon^2} = \frac{1}{n\varepsilon^2} \to 0 \text{ as } n \to \infty.$$

That means X_n is convergent in probability to 0. $\square$

Modes of Convergence

Example

Let $\{X_n\}_{i=1}^{\infty}$ be a sequence of discrete random variables. Let the probability function of a generic term X_n of the sequence be

$$p_{X_n}(x_n) = \begin{cases} 1/n & \text{for } x_n = n, \\ 1 - 1/n & \text{for } x_n = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Find the mean-square limit (if it exists) of the sequence $\{X_n\}_{i=1}^{\infty}$.

Solution. Note that

$$\lim_{n\to\infty} P(X_n = 0) = \lim_{n\to\infty} (1 - \frac{1}{n}) = 1.$$

Therefore, one would expect that the sequence $\{X_n\}$ converges to the constant random variable $X = 0$. However, the sequence does not converge in mean-square to $X = 0$. In fact, The distance of a generic term of the sequence from 0 is

$$E[(X_n - 0)^2] = n^2 \cdot \frac{1}{n} + 0 \cdot (1 - \frac{1}{n}) = n.$$

While if the possible values taken by X_n are $1/n$ and 0, then we would have

$$\lim_{n\to\infty} E[(X_n - 0)^2] = 0.$$

$\square$

Modes of Convergence

Example

Let X be a Bernoulli variable values 0 and 1 with equal probability $\frac{1}{2}$ and $X_1, X_2, \cdots$ identical random variables given by $X_n = X$ for all n. Then X_n are certainly not independent, but $X_n \xrightarrow{D} X$. Let $Y = 1 - X$. Clearly $X_n \xrightarrow{D} Y$ also, since X and Y have the same distribution. $\square$

Modes of Convergence

There are several relations among the various modes of convergence, which are summarized as following.

Theorem

The following implications hold:

$$(X_n \xrightarrow{a.s.} X) \Longrightarrow (X_n \xrightarrow{P} X) \Longrightarrow (X_n \xrightarrow{D} X),$$

$$(X_n \xrightarrow{r} X) \Longrightarrow (X_n \xrightarrow{P} X) \Longrightarrow (X_n \xrightarrow{D} X),$$

for any $r \geqslant 1$. *Also, if* $r > s \geqslant 1$ *then*

$$(X_n \xrightarrow{r} X) \Longrightarrow (X_n \xrightarrow{a.s.} X).$$

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The Law of Large Numbers

Let $\{X_n\}_{i=1}^{\infty}$ be a sequence of random variables with partial sums $S_n = \sum_{i=1}^{n} X_i$. We are interested in the asymptotic behavior of S_n as $n \to \infty$.
Indeed, let $S_n = X_1 + X_2 + \cdots + X_n$. The ***weak law of large numbers*** (WLLN) states that S_n converges to $E(S_n)$ in probability, i.e., for any $\varepsilon > 0$, we have

$$\lim_{n\to\infty} P\left(\left|\frac{S_n}{n} - E\left(\frac{S_n}{n}\right)\right| \geqslant \varepsilon\right) = 0.$$

Equivalently, the sample mean $\overline{X}_n = S_n/n$ converges to $E(\overline{X}_n)$ in probability, i.e., for any $\varepsilon > 0$, we have

$$\lim_{n\to\infty} P\left(|\overline{X}_n - E(\overline{X}_n)| \geqslant \varepsilon\right) = 0.$$

The Law of Large Numbers

While the ***strong law of large numbers*** (SLLN) says that $\overline{X}_n$ converges to $E(\overline{X}_n)$ almost surely, i.e., for any $\varepsilon > 0$, we have

$$P\left(\lim_{n\to\infty} |\overline{X}_n - E(\overline{X}_n)| = 0\right) = 1.$$

The Law of Large Numbers

In this section, we mainly introduce the weak law of large numbers.

Theorem

(***Chebyshev's law of large numbers***) *Let* $\{X_n\}_{i=1}^{\infty}$ *be an independent sequence, with finite expected value* $\mu_i = E(X_i)$. *If there exists a constant* $C > 0$ *such that* $Var(X_i) \leqslant C$, $i = 1, 2, \cdots$, *then for any* $\varepsilon > 0$,

$$P\left(\left|\overline{X}_n - \frac{\sum_{i=1}^{n} \mu_i}{n}\right| \geqslant \varepsilon\right) \to 0 \quad \text{as } n \to \infty.$$

Equivalently,

$$P\left(\left|\overline{X}_n - \frac{\sum_{i=1}^{n} \mu_i}{n}\right| < \varepsilon\right) \to 1 \quad \text{as } n \to \infty.$$

The Law of Large Numbers

Proof. Since $X_1, X_2, \cdots, X_n$ are independent, $E(\overline{X}_n) = (\sum_{i=1}^n \mu_i)/n$ and

$$Var(\overline{X}_n) = Var\Big(\sum_{i=1}^n X_n \Big/ n\Big) = \frac{1}{n^2}\sum_{i=1}^n Var(X_i) \leqslant \frac{C}{n}.$$

By Chebyshev's inequality, for any $\varepsilon > 0$,

$$P\left(\left|\overline{X}_n - \frac{\sum_{i=1}^n \mu_i}{n}\right| \geqslant \varepsilon\right) \leqslant \frac{Var(\overline{X}_n)}{\varepsilon^2} = \frac{C}{n\varepsilon^2}.$$

Thus, for fixed ε,

$$P\left(\left|\overline{X}_n - \frac{\sum_{i=1}^n \mu_i}{n}\right| \geqslant \varepsilon\right) \to 0 \quad \text{as } n \to \infty.$$

Equivalently,

$$P\left(\left|\overline{X}_n - \frac{\sum_{i=1}^n \mu_i}{n}\right| < \varepsilon\right) \to 1 \quad \text{as } n \to \infty.$$

$\square$

The Law of Large Numbers

As a special case of Chebyshev's law of large numbers, we have

Corralary

Let $X_1, X_2, \cdots, X_n$ be an IID sequence, with finite expected value $\mu = E(X_i)$ and finite variance $Var(X_i) = \sigma^2$. Then for any $\varepsilon > 0$,

$$P\left(\left|\overline{X}_n - \mu\right| \geqslant \varepsilon\right) \to 0 \quad \text{as } n \to \infty.$$

Note that $\overline{X}_n$ is an average of the individual outcomes, and one often calls the law of large numbers the "law of averages". It is a striking fact that we can start with a random experiment about which little can be predicted and, by taking averages, obtain an experiment in which the outcome can be predicted with a high degree of certainty.

The Law of Large Numbers

Consider the important special case of Bernoulli trials with probability p for success. Suppose that the number of successes in n trials is n_A. Let

$$X_i = \begin{cases} 1 & \text{if } A \text{ occurs at the } i\text{th trial,} \\ 0 & \text{otherwise.} \end{cases}$$

Then $n_A = X_1 + X_2 + \cdots + X_n$ and $\mu = E(X_i) = p$. Thus we have

Theorem

(***Bernoulli's law of large numbers***) *Suppose that n_A is the number of successes in n Bernoulli trials with probability p for success. Then for any $\varepsilon > 0$,*

$$P\left(\left|\frac{n_A}{n} - p\right| < \varepsilon\right) \to 1 \quad \text{as } n \to \infty.$$

The Law of Large Numbers

The above statement says that, in a large number of repetitions of a Bernoulli experiment, we can expect the proportion of times the event will occur to be near p. This shows that our mathematical model of probability agrees with our frequency interpretation of probability.

The Law of Large Numbers

Though we say probability always deals with random phenomenon, we can see an interesting example of solving a problem in calculus by using probability.

Example

Let $g(x)$ be a continuous function defined for $x \in [0,1]$ with values in $[0,1]$. Find the area $\int_0^1 g(x)dx$.

Solution. Here is a better way to estimate the same area. Let us choose a large number of independent values X_n at random from $[0,1]$ with uniform density, set $Y_n = g(X_n)$, and find the average value of the Y_n. Then this average is our estimate for the area. To see this, note that if the density function for X_n is uniform,

The Law of Large Numbers

$$\mu = E(Y_n) = \int_0^1 g(x)f(x)dx$$
$$= \int_0^1 g(x)f(x)dx = \text{average value of } g(x).$$

while the variance is

$$\sigma^2 = E\big[(Y_n - \mu)^2\big] = \int_0^1 (g(x) - \mu)^2 dx < 1,$$

since $g(x)$ is in $[0, 1]$ for all x in $[0, 1]$, μ is in $[0, 1]$. Hence $|g(x) - \mu| \leqslant 1$. Now let $\overline{Y}_n = (1/n)(Y_1 + Y_2 + \cdots + Y_n)$. From the law of large numbers, we see $\overline{Y}_n$ converges to the real area μ in probability. By Chebyshev's inequality, we have

$$P(|\overline{Y}_n - \mu| \geqslant \varepsilon) \leqslant \frac{\sigma^2}{n\varepsilon^2} < \frac{1}{n\varepsilon^2}.$$

The Law of Large Numbers

This says that to get within ε of the true value for $\mu = \int_0^1 g(x)dx$ with probability at least p, we should choose n so that $1/n\varepsilon^2 \leqslant 1 - p$ (i.e., so that $n \geqslant 1/(1-\varepsilon^2)p$). Note that this method tells us how large to take n to get a desired accuracy. $\square$

The Law of Large Numbers

From an actuarial point of view, Bernoulli's law of large numbers is considered to be the cornerstone and explains why and how insurance works. The main argument being that a pooling of similar uncorrelated risks $X_1, X_2, \cdots$ in an insurance portfolio S_N provides an equal balance within the portfolio that makes the outcome the more "predictable" the larger the portfolio size N is.

The Law of Large Numbers

Basically what this says is that for sufficiently large N there is only a small probability that the total claim S_N exceeds the threshold $(\mu + \varepsilon)N$, and thus, the bigger the portfolio the smaller the required security margin ε (per risk X_i) so that the total claim S_N remains below $(\mu + \varepsilon)N$ with sufficiently high probability. This is the foundation of the functioning of insurance; it is the aim of every insurance company to build sufficiently large and sufficiently homogeneous portfolios which makes the claim "predictable" up to a small shortfall probability.

The Central Limit Theorem

Suppose that X_1 has probability (density) function f. Then for all $n \geqslant 1$, the sample total $S_n = X_1 + X_2 + \cdots + X_n$ will have a density f_{S_n} (X_i are i.i.d., $i = 1, 2, \cdots, n$). Now $f_{S_1} = f$, and the other density can be calculated successively by using convolution formulas obtained in Chapter 3 for the density of the sum of two independent random variables. For certain choices of f (such as binomial, Poisson, normal), we can find simple formulas for f_{S_n}. In general, however, we have to resort to numerical methods.

The Central Limit Theorem

We have known that

$$E(S_n) = n\mu, \qquad \sigma_{S_n} = \sqrt{n}\sigma,$$

where $\mu = E(X_i)$, $\sigma^2 = Var(X_i)$.

But what is the distribution of S_n?

In fact, the interesting but most important thing is the shape of the sampling distribution approaches normal as the size of the sample increases, whatever the shape of the population distribution. To explain why many distributions tend to be close to the normal distribution, the central limit theorem (CLT) have been developed, which is the second fundamental theorem of probability.

The Central Limit Theorem

Definition

The standardized sum of S_n is given by

$$S_n^* = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

S_n^* always has expected value 0 and variance 1.

Theorem

(***Levy-Linderberg central limit theorem***) *Let $\{X_n\}_{n=1}^{\infty}$ be an IID sequence of random variables, each with mean μ and variance σ^2. Then the distribution F_n of S_n^* tends to the standard normal as $n \to \infty$. That is,*

$$F_n(x) = P(S_n^* \leqslant x) \overset{n\to\infty}{\longrightarrow} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-s^2/2} ds.$$

The Central Limit Theorem

Theorem

(***De Movire-Laplace central limit theorem***) *Suppose that* $X \sim B(n,\ p)$. *Then the distribution of*

$$\frac{X - np}{\sqrt{np(1-p)}}$$

tends to the standard normal as $n \to \infty$. *That is,*

$$P\left(\frac{X - np}{\sqrt{np(1-p)}} \leqslant x\right) \overset{n\to\infty}{\longrightarrow} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-s^2/2} ds.$$

The Central Limit Theorem

Proof. If X is binomially distributed with parameters n and p, then X has the same distribution as the sum of n independent Bernoulli random variables, each with parameter p. (Recall that the Bernoulli random variable is just a binomial random variable whose parameter n equals 1.) Hence, By Theorem 4, the distribution of

$$\frac{X - E(X)}{\sqrt{Var(X)}} = \frac{X - np}{\sqrt{np(1-p)}}$$

approaches the standard normal distribution as n approaches ∞. $\square$

The Central Limit Theorem

The normal approximation will, in general, be quite good for values of n satisfying $np(1-p) \geqslant 10$.

Example

(**Normal approximation to the Binomial**) Let X be the number of times that a fair coin, flipped 40 times, lands heads. Find the probability that $X = 20$. Use the normal approximation and then compare it to the exact solution.

Solution. Since the binomial is a discrete random variable and the normal a continuous random variable, it leads to a better approximation to write the desired probability as

The Central Limit Theorem

$$
\begin{aligned}
P(X = 20) &= P(19.5 < X < 20.5) \\
&= P\Big(\frac{19.5 - 20}{\sqrt{10}} < \frac{X - 20}{\sqrt{10}} < \frac{20.5 - 20}{\sqrt{10}}\Big) \\
&= P\Big(0.16 < \frac{X - 20}{\sqrt{10}} < 0.16\Big) \\
&\approx \Phi(0.16) - \Phi(0.16) \\
&= 2\Phi(0.16) - 1 = 0.1272.
\end{aligned}
$$

The exact result is

$$
P(X = 20) = \binom{40}{20}\left(\frac{1}{2}\right)^{40} = 0.1268.
$$

$\square$

The Central Limit Theorem

If $X_1, X_2, \cdots$ are not identical distribution, then we have

Theorem

(***Liapunov's central limit theorem***) *Suppose that* $\{X_n\}_{n=1}^{\infty}$ *are an independent random sequence such that* $E(X_n) = \mu_n$ *and* $Var(X_n) = \sigma_n^2 < \infty$. *Let* S_n *be sample total and*

$$S_n^* = \frac{S_n - E(S_n)}{\sigma_{S_n}} = \frac{S_n - \sum_{i=1}^{n} \mu_i}{\sqrt{\sum_{i=1}^{n} \sigma_i^2}}.$$

If there exist a constant $\delta > 0$, *such that*

$$\frac{1}{\sigma_{S_n^*}^{2+\delta}} \sum_{i=1}^{n} E\Big(|X_i - \mu_i|^{2+\delta}\Big) \to 0 \tag{1}$$

as $n \to \infty$, *then the distribution* F_n *of* S_n^* *tends to the standard normal as* $n \to \infty$. *That is,*

The Central Limit Theorem

That is,

$$F_n(x) = P(S_n^* \leqslant x) \ \rightarrow \ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-s^2/2} ds.$$

Equation (1) is called Liapunov condition.

The Central Limit Theorem

Example

Suppose that $X_i, i = 1, 2, \cdots, 12$ are uniform random variables in the interval $[0, 1]$. We shall determine the value of

$$P(|\overline{X}_{12} - \frac{1}{2}| \leqslant 0.1).$$

Solution. The mean of the uniform distribution in the interval [0,1] is 1/2 and the variance is 1/12. Since $n = 12$ in this example, the distribution of $\overline{X}_{12}$ will approximately a normal distribution with mean 1/2 and variance 1/144. Therefore, the distribution of the variable $Z = 12(\overline{X}_{12} - 1/2)$ will approximately a standard normal distribution. Hence,

$$P\left(\left|\overline{X}_{12} - \frac{1}{2}\right| \leqslant 0.1\right) = P\left(12\left|\overline{X}_{12} - \frac{1}{2}\right| \leqslant 1.2\right)$$

$$= P(|Z| \leqslant 1.2) = 2\Phi(1.2) - 1 \approx 0.7698.$$

□

The Central Limit Theorem

Example

The lifetime of a special type of battery is a random variable with mean 40 hours and standard deviation 20 hours. A battery is used until it fails, at which point it is replaced by a new one. Assuming a stockpile of 25 such batteries, the lifetimes of which are independent, approximate the probability that over 1100 hours of use can be obtained.

Solution. If we let X_i denote the lifetime of the ith battery to be put in use, then we desire $p = P(X_1 + \cdots + X_{25} > 1100)$. Since $E(X_i) = 40$ and $Var(X_i) = 20^2 = 400$, we get

$$E(X_1 + \cdots + X_{25}) = 1000, \quad \sigma_{X_1+\cdots+X_{25}} = \sqrt{400 \cdot 25} = 100,$$

The Central Limit Theorem

and the distribution of

$$\frac{X_1 + \cdots + X_{25} - 1000}{100}$$

tends to the standard normal as $n \to \infty$. That is,

$$\begin{aligned} p &= P\Big(\frac{X_1 + \cdots + X_{25} - 1000}{100} > \frac{1100 - 1000}{100}\Big) \\ &\approx P(N(0,1) > 1) = 1 - \Phi(1) \approx 0.1587. \end{aligned}$$

$\square$

The Central Limit Theorem

From the law of large numbers, we have known that if $\{X_n, n \geqslant 1\}$ is an IID sequence then $\overline{X}_n$ converges to $E(\overline{X}_n)$ in probability as $n \to \infty$. That is

$$P(|\overline{X}_n - E(\overline{X}_n)| \geqslant \varepsilon) \to 0 \quad \text{as } n \to \infty.$$

Now we will see how the probability converges to 0 exactly by using the central limit theorem.

$$\begin{aligned} P(|\overline{X}_n - E(\overline{X}_n)| \geqslant \varepsilon) &= 1 - P(|\overline{X}_n - E(\overline{X}_n)| < \varepsilon) \\ &= 1 - P\Big(\Big|\frac{\overline{X}_n - E(\overline{X}_n)}{\sqrt{Var(\overline{X}_n)}}\Big| < \frac{\varepsilon}{\sqrt{Var(\overline{X}_n)}}\Big) \\ &= 2(1 - \Phi(\sqrt{n}\varepsilon/\sigma)), \end{aligned}$$

where σ is the standard deviation of X_i.

The end

Thank you !